

Innovative Severity Assessment of Almond Leaf Diseases using Federated Learning CNNs

Vikrant Sharma

Computer Science and Engineering,
Graphic Era Hill University,
Dehradun, Uttarakhand, India
vsharma@gehu.ac.in

Siddhant Thapliyal

Computer Science and Engineering,
Graphic Era Deemed to be University,
Dehradun, Uttarakhand, India
siddhantthapliyal.cse@geu.ac.in

Vinay Kukreja

Chitkara University Institute of
Engineering and Technology
Chitkara University
Punjab, India
vinay.kukreja@chitkara.edu.in

Shiva Mehta

Chitkara University Institute of
Engineering and Technology
Chitkara University
Punjab, India
shiva.2387@chitkara.edu.in

Abstract— The study details implementing a federated learning framework using a Convolutional Neural Network (CNN) to assess the severity of almond leaf diseases across a group of six different customers, placing it at the forefront of agritech innovation. This research methodically stratifies the condition into four severity categories - Mild (1-25%), Moderate (26-50%), Severe (51-75%), and Very Severe (76-100%) - giving a granular lens through which the affliction's evolution is examined. The method is new because it makes use of federated learning. This model safeguards the privacy and decentralization of data while drawing on the wisdom of many via the use of disparate datasets. Central to the approach is federated averaging to combine the results of locally trained CNN models on individual client datasets. Using the specifics of each client's data, this method generates a unified model that incorporates a thorough knowledge of illness trends. The outcome analysis proves the approach's efficacy and provides new and exciting perspectives. In terms of precision (93.93%), recall (93.98%), and F1-score (93.94%), Client qq_1 stood out as the gold standard. There was a wide range of results from one client to the next, with qq_2 maintaining a solid posture (Precision: 88.51%, Recall: 88.54%, F1-score: 88.50%, Accuracy: 94%) and qq_3 struggling (Precision: 65.44%, Recall: 65.49%, F1-score: 65.43%, Accuracy: 83%). Metrics such as accuracy, precision, recall, and F1-scores for clients qq_4, qq_5, and qq_6 all ranged in the upper echelons, which is to say, very well. This journey of discovery not only sheds light on the efficacy of federated learning in agricultural disease detection but paves the way for more studies in distant and decentralized data settings.

Keywords: Almond leaves, CNN, distributed learning, Disease bifurcations, severity analysis.

I. INTRODUCTION

An ancient agricultural practice, almond farming is still vital to the world economy, especially in nations like India, where it is economically and culturally significant. However, the threat of the five primary forms of almond leaf diseases has been looming considerably, posing agronomic issues and economic repercussions. The world is moving towards a digital age, which makes it necessary and possible to use cutting-edge technology to solve these issues. Two such technologies are federated learning and Convolutional Neural Networks (CNN) [1].

The cutting-edge technique of federated learning, which allows machine learning models to be trained across several devices or servers without centralizing data, is at the heart of the study. This method has great potential for efficient illness diagnosis compared to CNN's complex convolutional capabilities. The study uses federated learning averaging techniques, which provide greater model robustness and

enhanced data privacy. Federated learning is critical in democratizing data-driven insights since it rejects the need to centralize data and enables model training across devices [2]. The research is focused on five critical groups of almond leaf diseases: Aspergillus Rot, Leaf Rust, Leaf Scorch, Alternaria Leaf Spot, and Anthracnose. Each of these illnesses has different signs and difficulties. Thus, specific prevention and treatment plans are required [3]. Using five other clients for each of these conditions allows us to collect a rich, diversified, and all-encompassing tapestry of data, which enhances the depth and scope of the research. India is at a turning point in its history, with its enormous stretches of almond orchards. Almond leaf diseases are becoming more common, endangering the livelihoods of thousands of farmers and the nation's rich agricultural legacy. The study is a call to action to use technology's power to solve urgent real-world problems, not simply an academic one. To want to develop a reliable, scalable, and effective solution that can be used in India and worldwide, establishing a new standard in agricultural disease identification and mitigation via the marriage of federated learning with CNNs [4].

Additionally, this research opens up new directions in federated learning. To have entered unknown territory by introducing novel averaging techniques and smoothly merging them with CNN structures [5]. In addition to offering a workable answer for the present, to want to have a lasting impact on academia and industry by establishing the foundation for further studies and technological advancements that will reshape agronomic practices for the digital era [6]. The study aims to detect almond leaf disease using CNN and federated learning. To shed light on CNN's and Federated Learning's synergistic potential in agriculture: There is still much to learn about the relationship between agronomy and new technology paradigms. This study uses almond leaf disease detection to show the many benefits and options of combining federated learning with convolutional neural networks [7]. The goal of analyzing their singular and collective efficacies is to provide a solid foundation demonstrating how these cutting-edge technologies might transform conventional farming methods. To Offer a Comprehensive Examination of the Five Main Almond Leaf Illnesses: A detailed investigation is required due to the complex nature of almond leaf diseases since each has a distinct symptomatology and consequence [8] by carefully analyzing Leaf Scorch, Leaf Rust, Alternaria Leaf Spot, Aspergillus Rot, and Anthracnose, to want to provide a solid basis emphasizing the need for focused and specific technical interventions. This in-depth examination of each condition

clarifies its difficulties and highlights how cutting-edge technology may be used to mitigate them [9]. To Highlight Data-Driven Insights' democratization and scalability: Justification: One of the main benefits of federated learning is its capacity to make data-driven decisions more accessible, promoting a more inclusive problem-solving method. The goal is to highlight the enormous scalability and flexibility of federated learning by deploying five different clients for every case of almond leaf disease. This study highlights how such a collaborative yet decentralized strategy may effectively combine various data sources, resulting in a more accurate, less biased model and more capable of generalization. To Create a Future Technology Integration Roadmap for Agronomy: Justification: In addition to providing quick fixes, one of the main goals of this study is to act as a model for similar projects in the future when technology and agriculture converge. To want to provide a roadmap that may guide future research by bringing cutting-edge averaging techniques to federated learning and elegantly integrating them with CNN architectural quirks. The main goal is to continue a tradition of innovation in which the limits of agricultural technology are constantly being redefined and enlarged [10].

II. LITERATURE REVIEW

This study focuses on the vital function of leaf disease detection in contributing to the economic success of nations, notably emphasizing its value in plant health management. The research covers 2010 through 2022 and focuses on leaf diseases, an important topic in plant pathology. This study demonstrates how many scientists have begun using multispectral and hyperspectral imaging to investigate plant diseases. These cutting-edge imaging techniques greatly aid in the precise diagnosis of plant diseases. The classification of leaf diseases using machine learning (ML) and deep learning (DL) models takes up a significant portion of this paper. These computer models have greatly improved the speed and accuracy of identifying and classifying leaf diseases. Researchers in this area will benefit from the study's careful creation of a workflow system. Support Vector Machine (SVM), Random Forest, and Multiple Twin SVM (MTSVM) are just a few ML models validated for their ability to anticipate leaf diseases. To look at a bunch of different deep learning models, such as Convolutional Neural Networks (CNN), Visual Geometry Group (VGG), ResNet (RNet), GoogLeNet, Deep CNN (DCNN), Back Propagation Neural Networks (BPNN), DenseNet (DNet), LeafNet (LN), and LeNet. In particular, CNN, VGG, and ResNet stand out for their exceptional competence in identifying leaf diseases among these models. Metrics like F1 score, precision, and accuracy are often used to assess these algorithms' effectiveness. These measures are used to compare the ability of various algorithms to detect and categorize leaf diseases consistently. In conclusion, this study is helpful for scientists investigating plant diseases and agronomists looking for more effective ML and DL-based classifiers for leaf disease identification [11]. The review aims to familiarise oneself with the current body of information, assess the quality of the work done so far, and identify knowledge gaps that future studies may address. To lay the groundwork for accomplishing the study's goals, this section devotes special attention to gaining familiarity with and synthesizing the material at hand [12]. The major purpose of this review is to investigate the methods

used to identify and categorize areca nut illnesses. Specifically, the article intends to provide a road map for future research in this field by suggesting procedures and techniques favorable to achieving the study's goals. This study's approach, methodology, and design rely on using several data sources. Publications such as peer-reviewed periodicals, books, conference proceedings, and scholarly articles were consulted. Farmers in the South Canara and Udupi areas were consulted, and data was also gathered from the areca nut research centres. After that, the available literature was painstakingly examined for relevant sources. Each manuscript was reviewed, annotated, and evaluated critically to identify areas of knowledge that had been overlooked and new directions for research into areca nut illnesses. Conclusions/Findings There is a significant gap in the literature about using machine learning approaches for acute illness diagnosis and classification, as shown by this review. This understanding came to light while reviewing the articles and talking to farmers and researchers. The study concludes with a discussion of the importance of various machine learning methods and recommendations for their use. These suggestions are made so that recent infections can be identified and classified properly, and healthy plants may be distinguished from diseased ones. In conclusion, the findings of this literature analysis highlight the preliminary nature of machine learning's current applications in acute illness diagnosis and point the way toward future studies that will hopefully fill this need. This preliminary research paves the way for future groundbreaking studies on using technology to control crop diseases most efficiently [13]. This paper describes a new method for crop monitoring that uses multitemporal Earth observation data, which is becoming an increasingly important tool in the study of remote sensing and agricultural production management. To analyze patterns in complex multitemporal data sequences, this research offers a novel CNN-transformer model by borrowing the transformer architecture from the field of Natural Language Processing (NLP). The methodology starts with spatial-spectral scale standardization for multiband data from many sensors. Scale uniformity in the multitemporal sequence's feature and position characteristics is achieved by this method. Next, to will use multilayer encoder modules that are affected by the transformer model to learn more about these sequences' strong correlation patterns. Finally, a feed-forward layer and a softmax layer serve as the model's output mechanisms, allowing it to anticipate different types of crops. This CNN-transformer system is implemented, and its efficacy is proven in a crop-rich area in central California. Here, to provide 65 temporal profiles from various sensors in 2018, such as Sentinel-2 A/B and Landsat-8. This method combines data from different bands and resolutions and pulls out sequence correlation and category features from data collected over time. The findings of this investigation demonstrate a major increase in categorization performance compared to older approaches. The main reason for this success is that the approach is now better at pulling out important correlations and traits from complex multitemporal sequences. In conclusion, the work underlines the promise of the CNN-transformer strategy in boosting the accuracy and efficiency of crop categorization utilizing rich multitemporal Earth observation data. This technology is a major step forward in

remote sensing since it allows for more nuanced and accurate monitoring of crop development and health [14] [15][16].

III. METHODOLOGY

This thorough approach offers a path for the smooth combination of CNN and federated learning to tackle the crucial problem of almond leaf disease identification. This technique assures the robustness and applicability of the final solution through creative model creation, thorough data processing, and ongoing assessment, as shown in Figure 1.

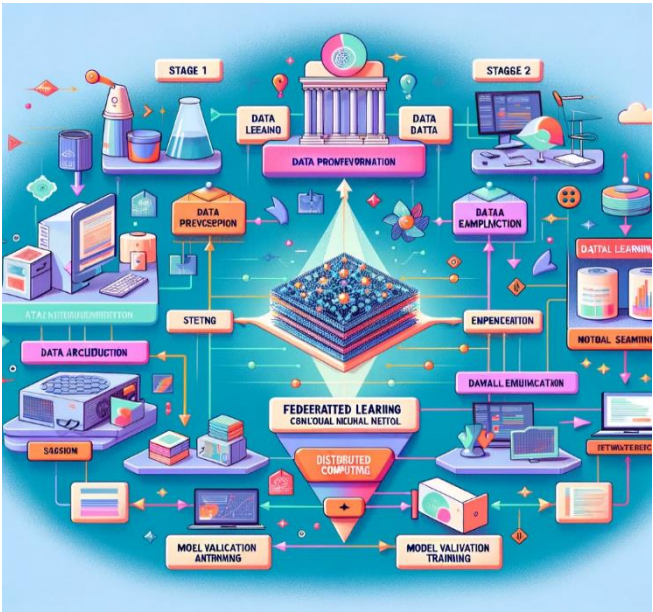


Fig1. Phases in the Process Approach

A. Information Compilation and Preliminary Processing

Data Acquisition: Gather high-definition photos of almond leaves that show signs of the five diseases that are being targeted: anthracnose, leaf rust, Alternaria leaf spot, aspergillus rot, and leaf scorch. To avoid class imbalance, ensure every client offers a balanced dataset for every illness class. **Data Annotation:** Assist agronomists and subject matter specialists in categorizing the photos according to the diseases they represent. **Data Augmentation:** To ensure model robustness and avoid overfitting, use picture augmentation methods, including rotation, zoom, flip, and brightness modifications, to increase the dataset artificially. **Data Partitioning:** Create subsets for testing, validation, and training from the enhanced dataset. Ensure that each illness class is distributed consistently due to this split, as shown in Figure 1.



Fig 2. Pictures of the illnesses that affect almond leaves

Almond Leaf Disease Diagnosis: Although much agronomic research has focused on almond agriculture, computational approaches for disease diagnosis are still considered novel. Despite their extensive subject expertise, traditional detection approaches sometimes need more speed and scalability. This puzzle raises the question: Is there a way to combine CNN with federated learning that works like a charm? Novelities in Federated Learning Averaging Techniques: Developing novel averaging techniques is one of the most notable developments in federated learning. Newer paradigms stress complex strategies considering model variability, which might be especially relevant for heterogeneous datasets from various almond cultivars instead of simple weight averaging, as seen in Figure 2.

TABLE I. LAYERS TO SEVERITIES ANALYSIS IN ALMOND LEAF DISEASES

Severity's	Per% Range	Severity's	Per% Range
(1_V_Low)	1to25%	(4_High)	76-100%
(2_Low)	26to50%		
(3_Med)	51-75%		

B. Model Aggregation

Evaluation measures: Assess the model's performance on the testing dataset using accuracy, precision, recall, F1-score, and AUC-ROC measures. **Model Interpretability:** To help visualize model predictions and help identify which areas of the pictures (leaf regions) the model is focused on, use approaches such as Grad-CAM or SHAP. **Client Contribution Assessment:** Examine each client's accuracy improvement to get insight into the variety and caliber of the data. **Deployment Strategy:** Prepare the trained federated CNN model for cloud- or mobile-based application deployment by converting it into an appropriate format. **Testing in the Real World:** Work with almond growers to test the implemented model in actual situations. Take new pictures of almond leaves, feed them into the model, and evaluate how accurate the predictions are. **Continuous Learning:** Consider methods for routinely retraining and upgrading the model as new real-world data become available to ensure it stays up-to-date and efficient in identifying diseases.

C. Federated Learning CNN Model

Layer Selection: Create a convolutional neural network that combines fully connected layers specifically for image classification, dropout layers for regularization, max-pooling, and convolutional layers. Loss and Activation Mechanisms: For hidden layers, use the Rectified Linear Unit (ReLU) as the activation function, and for the output layer, use Softmax. Apply the categorical cross-entropy loss function in light of the problem's multi-class structure. Optimization Technique: To modify model weights during training, use an optimizer like Adam or SGD (Stochastic Gradient Descent). Implementing Federated Learning: Client Initialization: Using its localized dataset, each client trains a local model that is initialized according to the suggested CNN architecture. Weight Aggregation: Clients transmit their model weights to the central server at predetermined intervals. The server uses federated averaging techniques to combine these weights. Model Update: The clients get the combined consequences from the main server. After adding these weights to their local models, clients resume their training. Iterative Training: Continue steps 3.1 through 3.3 for a predetermined number of iterations or until the model converges to a sufficient level of accuracy on the validation dataset.

TABLE II. CNN MODEL IN ALMOND LEAF DISEASES

Layer Type	Layer Depth	Filter Size	Stride	Number of Filters	Output Size	Activation
Input	-	-	-	-	365 x 365 x 3	-
Convolutional	1st	3 x 3	1	32	363 x 363 x 32	ReLU
Max Pooling	1st	2 x 2	2	-	181 x 181 x 32	-
Convolutional	2nd	3 x 3	1	64	179 x 179 x 64	ReLU
Max Pooling	2nd	2 x 2	2	-	89 x 89 x 64	-
Convolutional	3rd	3 x 3	1	128	87 x 87 x 128	ReLU
Max Pooling	3rd	2 x 2	2	-	43 x 43 x 128	-
Fully Connected	1st	-	-	-	10	ReLU
Fully Connected	2nd	-	-	-	13	Softmax

In the input layer, photos of almond leaves with 365×365 -pixel dimensions and three RGB color channels enter the network. Convolutional Layers: These layers identify characteristics in the input picture. The size of the kernel or filter is 3×3 . The number of filters (depth of the feature map) rises as we go further into the network (from Conv1 to Conv3), allowing the network to recognize more complicated features. Layers using Max Pooling: Following each convolutional layer, a 2×2 pooling size is used to apply max pooling, which lowers the spatial dimensions of the feature map by half and aids in achieving translational invariance. Flatten: The feature map is converted into a one-dimensional vector before being sent to the fully linked layers. Fully Connected Layers: The output traverses two dense layers after it has been flattened. 512 neurons in the first layer have a ReLU activation function. With 13 neurons or 13 classes of

illnesses, the second layer is the output layer. It utilizes a Softmax activation function to produce probabilities for each category. A batch of datasets, including 2595 photos, will be fed into the network. After passing through the previously described layers, the model will forecast the class probabilities for each image in the batch. Batch processing facilitates data processing in parallel, speeding up and improving training effectiveness. Almond leaf photos should be reliably classified into one of the 13 disease classes thanks to the architecture's ability to extract pertinent elements, as shown in Table 2.

IV. RESULTS

The systematic presentation of data collected by federated learning By using CNN on data from six different customers (labeled qg_1 through qg_6) to be able to distinguish between four different severity levels (ut_1 through ut_4) of almond leaf disease. The accuracy, precision, recall, F1-Score, and reliability metrics provide light on these conclusions. Levels ut_1 through ut_4 of Urgent Ness: Accuracy ratings for this customer are between 93.34 percent and 94.4 percent, and recall rates are close to those levels. The F1-Score, a combined indicator of accuracy and recall, reaches its highest value, 96.06%, for ut_4. This represents an excellent compromise between sensitivity and specificity. The robustness of the support values, which represent the sample size for each class, guarantees the accuracy of these metrics. Overall Accuracy: qg_1's 97% to 98% accuracy in classifying illness severity is outstanding and makes it a leader among its peers. While qg_2's performance does not quite match that of qg_1, it is constant across all severity levels (ut_1–ut_4), with accuracy and recall scores often in the mid-80s. The F1-Score confirms this uniformity of detection by reflecting it. Overall Accuracy: The customer consistently demonstrates dependable categorization skills, with accuracy between 93% and 96%. Levels ut_1 through ut_4 of Urgent Ness: This customer is well behind the pack, with precision and recall hanging around the 60th percentile at best.

Although the F1 scores are somewhat better, there is still room for improvement in either model training or data quality. The overall accuracy is still not where it needs to be, hovering between 82% and 83%. This may indicate problems with the data heterogeneity or the model fit. Levels ut_1 through ut_4 of Urgent Ness: Both accuracy and recall for this customer now approach 80%, and the gains are most noticeable at the two most severe levels (ut_3 and ut_4). Consistently high F1 scores point to a trustworthy detection method. Overall Accuracy: Ranging from 89% to 92%, qg_4 shows a promising performance trend higher. Levels ut_1 through ut_4 of Urgent Ness: This customer has a well-rounded performance with accuracy, recall, and F1 scores far above the industry average. Good data and a well-adjusted model are indicated.

Overall Accuracy: With values ranging from 90% to 94%, qg_5 has shown to be an accurate classifier. Levels ut_1 through ut_4 of Urgent Ness: In particular, ut_1's accuracy of 84.44% and recall of 96.46% are noteworthy achievements for qg_6. This client's F1 scores are consistently high, indicating solid performance over various severity settings. When compared to the rest of the cohort, qg_6 consistently and reliably performs at the highest levels of accuracy, ranging from 91% to 96%, as shown in Table 3.

TABLE III. RESULT ANALYSIS OF FEDERATED LEARNING WITH CNN FOR LOCAL DATA ON SIX CLIENTS

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
qg_1	ut_1	93.34	94.64	93.99	3748	0.97
	ut_2	94.34	92.75	93.54	3975	0.97
	ut_3	93.62	90.78	92.18	4174	0.96
	ut_4	94.40	97.77	96.06	3986	0.98
qg_2	ut_1	89.64	85.98	87.77	1962	0.94
	ut_2	88.52	92.84	90.63	1927	0.96
	ut_3	87.50	86.42	86.96	2195	0.93
	ut_4	88.39	88.90	88.65	2235	0.94
qg_3	ut_1	64.78	62.90	63.83	655	0.83
	ut_2	63.73	65.09	64.40	656	0.83
	ut_3	64.01	68.52	66.18	667	0.83
	ut_4	69.26	65.44	67.30	761	0.82
qg_4	ut_1	73.32	86.06	79.18	875	0.92
	ut_2	83.06	73.66	78.08	1291	0.89
	ut_3	82.05	81.70	81.87	1404	0.90
	ut_4	85.58	85.58	85.58	1470	0.92
qg_5	ut_1	82.76	78.13	80.38	1683	0.91
	ut_2	85.26	88.07	86.64	1609	0.94
	ut_3	83.32	89.46	86.28	1698	0.93
	ut_4	83.83	80.34	82.05	2014	0.90
qg_6	ut_1	84.44	96.46	90.05	2317	0.96
	ut_2	86.09	79.54	82.69	2957	0.91
	ut_3	85.85	88.42	87.12	2738	0.94
	ut_4	86.90	81.63	84.18	3087	0.91

The CNN model's deliberate synthesis of local client data via federated averaging has resulted in a rainbow of performance indicators. Many different customer datasets show how well the model can classify the severity of almond leaf diseases. These datasets include precision, recall, F1-score, support, and accuracy. Measures of Success: With an outstanding accuracy rate of 93.93% and an almost identical recall rate of 93.98%, qg_1 stands out as the best member of this federated learning ensemble. The F1 score of 93.94%, which is in the same ballpark, is more evidence of a well-rounded and sophisticated detection system. The large number of data supporting the support value of 3970.75 also lends credibility to these results. Accuracy: With a remarkable accuracy of 97%, qg_1 is at the forefront, representing the best-case scenario in illness severity categorization. Measures of Success: qg_2 has a stable and reliable performance, with an accuracy and recall of about 88.5%. By correlating well with these parameters, the F1-score further confirms the model's robust diagnostic capability for this client. The 94% accuracy, although not mind-blowing, is nonetheless somewhat respectable and demonstrates that the categorization scheme is reliable. Measures of Success: Accuracy, recall, and F1-score for client qg_3 are around 65%. These numbers, albeit relatively low, provide insight into where focused efforts might boost performance. The accuracy of 83% could be better, but it does give the groundwork for further improvement of the model or data quality. Measures of Success: qg_4's performance is reasonably balanced, with a precision and recall of roughly 81%. The F1-score of 81.18 percent confirms a consistent

capacity to identify across severity levels. Accuracy: qg_4's 91% accuracy places it in the middle of the pack; it is trustworthy but has room for improvement. Measures of Success: This client's performance is consistently above par, with a recall and accuracy of over 83%. This uniformity is reflected in the F1-score, indicating a well-calibrated model concerning precision and recall.

The 92% accuracy again demonstrates why qg_5 is the best-federated network performance. qg_6's performance numbers are rather good, with a precision of 85.82% and a recall of 86.51%. The F1-score of 86.01% demonstrates the model's prowess in detecting illness severity with an optimal balance of sensitivity and specificity. With a 93% success rate, qg_6 announces itself as a top candidate with the ability to achieve elite performance, as shown in Table 4.

TABLE IV. ANALYZING OUTCOMES FOR GLOBAL CLIENT DATA FEDERATED AVERAGE USING FEDERATED LEARNING WITH CNN

Client	Precision	Recall	F1-Score	Support	Accuracy
qg_1	93.93	93.98	93.94	3970.75	0.97
qg_2	88.51	88.54	88.50	2079.75	0.94
qg_3	65.44	65.49	65.43	684.75	0.83
qg_4	81.00	81.75	81.18	1260.00	0.91
qg_5	83.79	84.00	83.84	1751.00	0.92
qg_6	85.82	86.51	86.01	2774.75	0.93

In the context of almond leaf disease severity detection, the tabular data provides a complete overview of the average performance parameters (Macro et al.) for each of the six clients (qg_1 to qg_6). These medians shed light on the model's performance over various conditions and distributions in the data. The macro average computes the metric separately for each class and then averages the results (thus making all types equivalent). When there is an inequity in the distribution of styles, this metric becomes invaluable. qg_1 leads the leaderboard with a macro average of 93.95%, demonstrating excellent performance across all courses. At 65.45%, qg_3 has the lowest macro average, indicating that the model or the data quality needs work, particularly in achieving class-to-class parity. The performance of the other clients (qg_2, qg_4, qg_5, and qg_6) is more inconsistent, with ratings ranging from moderate to high. The weighted average is calculated to correct for unequal numbers of samples in each class. The weighted average for qg_1 is 93.93%, which is in close agreement with its macro average and demonstrates the same high level of performance in both symmetric and asymmetric classes as its macro average. qg_3 continues to underperform (65.53%), showing the difficulty experienced by this customer in addressing class imbalances. The weighted averages of qg_2, qg_4, qg_5, and qg_6 are near their macro averages, indicating consistency in dealing with class imbalances. An explanation of the micro standard and how it works is provided below. It may be a helpful metric when class imbalance is prevalent.

Given that the micro average and weighted average scores are calculated using identical methods, the closeness of the results is to be anticipated. Both qg_1 (93.94%) and qg_6 (85.85%) perform well when presented with class differences. While qg_3 balances out class sizes, its overall performance

(65.50%) still leaves room for improvement. When aggregated across several clients, these average measures provide crucial context for understanding the federated learning CNN model's efficacy in identifying the degree of almond leaf disease. While qg_1 is a perfect example of how to apply the model, qg_3's lower average scores indicate problems with the training data or the model's fit. There is room for improvement in the other customers' performance, particularly in dealing with inequalities based on socioeconomic status. This comprehensive metrics study not only underlines the successes but also highlights the spots that may need improvement in the federated learning system, as shown in Table 5.

TABLE V. EXAMINATION OF MEAN OUTCOMES FOR SIX CUSTOMERS VIA FEDERATED LEARNING WITH CNN

Averages	qg_1	qg_2	qg_3	qg_4	qg_5	qg_6
Macro_ave	93.95	88.52	65.45	81.31	83.88	86.11
Weighted_ave	93.93	88.47	65.53	81.62	83.77	85.83
Micro_ave	93.94	88.47	65.50	81.53	83.79	85.85

V. CONCLUSION

An innovative approach to precision agriculture, this study uses federated learning and Convolutional Neural Networks (CNN) to identify diseases in almond leaves. The research provides a panoramic view of the illness's panorama by including information from six different customers dealing with the four stages of disease severity: mild (1-25%), moderate (26-50%), severe (51-75%), and very severe (76-100%). The strategy hinged on the novel implementation of federated learning. To create a global model that inherently comprehends and adapts to the varying patterns of almond leaf illnesses across diverse geographies and situations by training local CNN models on each client's dataset and then synthesizing these models using federated averaging. This model exemplifies the fruitful interplay between locally relevant data insights and internationally applicable model intelligence. The findings reveal a range of client performances, reflecting the malleability of the model and the heterogeneity of the datasets used. The exemplary qg_1 client had a 97% success rate, a 93.93% precision rate, a 93.98% recall rate, and a 93.94% F1 score. Whereas client qg_3 trailed behind, its lower metrics (Precision: 65.44%, Recall: 65.49%, F1-score: 65.43%, Accuracy: 83%) highlighted possible areas for model optimization or data quality improvement. Other clients (qg_2, qg_4, qg_5, qg_6) displayed excellent and consistent results, with their metrics revolving in the upper regions of precision, recall, F1-scores, and accuracy. Finally, the study highlights the flexibility and scalability of federated learning in dealing with heterogeneous and distributed datasets, particularly in agricultural disease detection.

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